EXP 801 DS Wheat Heat Drought

— Ed Chaplin (USyd), *Apr–June 2025* —

Li-600 STOMATAL CONDUCTANCE MODEL PARAMETER ESTIMATION - LIGHT OFF TO LIGHT ON (INCREASE)

- Using the final model fitted to the data, obtain the parameters of interest
- These include:
- k_i, being the time constant for gs to increase to Gsmax
- lambda, being the lag in the response time of gs to a step increase in light
- g0, being the starting value of the gs response (intercept/lower asymptote)
- g1, being the upper asymptote of the gs response (this will be used to calculate iWUE)
- Slmax, a derived trait from McAusland paper
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Load the libraries

```
library(knitr)
#knitr::spin("Exp801.porometer.stomatalCondParamEstimation_i.v1.r")
library(ggplot2)
opts_chunk$set("tidy" = FALSE, "dev" = "png", "dpi" = 300,
               "fig.width" = 7.5, "fig.height" = 10, "comment"=NA, fig.path = "figures/increase/")
library(dae)
library(dplyr)
library(brms) ##package used to fit bayesian regression model using Stan
library(tidyr)
library(emmeans)
library(tidybayes)
```

Set up constants and files

```
options(width=90)
load("../data/porometer.anal_i.v1.Rdata")
saveFile <- "../data/stomatalCondParamEstimation_porometer_i.v1.Rdata"
```

Have a look at the final model summary and store fixed effects

```
summary(fit_on_final)
```

```
 Family: gaussian
  Links: mu = identity
Formula: gsw ~ g0 + (g1 - g0) * exp(-exp(((lambda - time.adj)/k) + 1))
         g0 ~ 1 + Line * Device_Watering * Room_Temperature + (1 | CEBRoom:CEBSide:CEBLocation + CEBRoo
         g1 ~ 1 + Line + Device_Watering + Room_Temperature + Line:Device_Watering + Device_Watering:Ro
         k ~ 1 + Line
         lambda ~ 1
   Data: li600.on.dat (Number of observations: 1682)
  Draws: 6 chains, each with iter = 10000; warmup = 6000; thin = 1;
         total post-warmup draws = 24000

Multilevel Hyperparameters:
~CEBRoom:CEBSide:CEBLocation (Number of levels: 8)
                 Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
```

```
sd(g0_Intercept)     0.02      0.02     0.00     0.06 1.00     6963    11013
sd(g1_Intercept)     0.03      0.03     0.00     0.10 1.00     9505    13168

~CEBRoom:CEBSide:CEBLocation:CEBSubUnit (Number of levels: 16)
                 Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sd(g0_Intercept)     0.05      0.01     0.02     0.08 1.00     7495    10063
sd(g1_Intercept)     0.09      0.03     0.06     0.15 1.00     6857    11296

~CEBRoom:CEBSide:CEBLocation:CEBSubUnit:CEBCell (Number of levels: 159)
                 Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sd(g0_Intercept)     0.01      0.01     0.00     0.02 1.00     9583    11234
sd(g1_Intercept)     0.09      0.01     0.08     0.11 1.00     7107    13229

Regression Coefficients:
                                                                                                  Estimate
g0_Intercept                                                                                          0.01
g0_LinePBI17N002M0CM0NM0NM010NM16N                                                                   -0.00
g0_LinePBI17N006M0NM0NM89N                                                                           -0.00
g0_LinePBI17N007M0CM0NM0NM010NM8N                                                                     0.02
g0_LinePBI17N013M0CM0NM0NM010NM8N                                                                     0.02
g0_LinePBI17N016M0CM0NM0NM010NM27N                                                                   -0.01
g0_LinePBI19N007M0NM3N                                                                               -0.01
g0_LinePBIC15034M0CM24NM010NM1NM0N                                                                   -0.01
g0_LineROCKSTAR                                                                                      -0.01
g0_LineSUNCHASER                                                                                      0.01
g0_Device_WateringD2_Limited                                                                          0.03
g0_Room_TemperatureCER8_Ambient                                                                      -0.01
g0_LinePBI17N002M0CM0NM0NM010NM16N:Device_WateringD2_Limited                                         -0.02
g0_LinePBI17N006M0NM0NM89N:Device_WateringD2_Limited                                                 -0.01
g0_LinePBI17N007M0CM0NM0NM010NM8N:Device_WateringD2_Limited                                          -0.01
g0_LinePBI17N013M0CM0NM0NM010NM8N:Device_WateringD2_Limited                                           0.01
g0_LinePBI17N016M0CM0NM0NM010NM27N:Device_WateringD2_Limited                                         -0.00
g0_LinePBI19N007M0NM3N:Device_WateringD2_Limited                                                     -0.01
g0_LinePBIC15034M0CM24NM010NM1NM0N:Device_WateringD2_Limited                                          0.04
g0_LineROCKSTAR:Device_WateringD2_Limited                                                            -0.01
g0_LineSUNCHASER:Device_WateringD2_Limited                                                            0.03
g0_LinePBI17N002M0CM0NM0NM010NM16N:Room_TemperatureCER8_Ambient                                       0.00
g0_LinePBI17N006M0NM0NM89N:Room_TemperatureCER8_Ambient                                               0.00
g0_LinePBI17N007M0CM0NM0NM010NM8N:Room_TemperatureCER8_Ambient                                       -0.01
g0_LinePBI17N013M0CM0NM0NM010NM8N:Room_TemperatureCER8_Ambient                                        0.01
g0_LinePBI17N016M0CM0NM0NM010NM27N:Room_TemperatureCER8_Ambient                                      -0.00
g0_LinePBI19N007M0NM3N:Room_TemperatureCER8_Ambient                                                  -0.01
g0_LinePBIC15034M0CM24NM010NM1NM0N:Room_TemperatureCER8_Ambient                                       0.01
g0_LineROCKSTAR:Room_TemperatureCER8_Ambient                                                          0.00
g0_LineSUNCHASER:Room_TemperatureCER8_Ambient                                                        -0.02
g0_Device_WateringD2_Limited:Room_TemperatureCER8_Ambient                                             0.00
g0_LinePBI17N002M0CM0NM0NM010NM16N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient            -0.01
g0_LinePBI17N006M0NM0NM89N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient                     0.00
g0_LinePBI17N007M0CM0NM0NM010NM8N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient            -0.00
g0_LinePBI17N013M0CM0NM0NM010NM8N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient             0.01
g0_LinePBI17N016M0CM0NM0NM010NM27N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient           -0.00
g0_LinePBI19N007M0NM3N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient                        -0.00
g0_LinePBIC15034M0CM24NM010NM1NM0N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient             0.01
g0_LineROCKSTAR:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient                                -0.00
```

g0_LineSUNCHASER:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	-0.01
g1_Intercept	0.23
g1_LinePBI17N002M0CM0NM0NM010NM16N	0.01
g1_LinePBI17N006M0NM0NM89N	0.09
g1_LinePBI17N007M0CM0NM0NM010NM8N	0.08
g1_LinePBI17N013M0CM0NM0NM010NM8N	0.05
g1_LinePBI17N016M0CM0NM0NM010NM27N	-0.02
g1_LinePBI19N007M0NM3N	0.13
g1_LinePBIC15034M0CM24NM010NM1NM0N	0.10
g1_LineROCKSTAR	0.11
g1_LineSUNCHASER	0.09
g1_Device_WateringD2_Limited	-0.14
g1_Room_TemperatureCER8_Ambient	0.01
g1_LinePBI17N002M0CM0NM0NM010NM16N:Device_WateringD2_Limited	0.12
g1_LinePBI17N006M0NM0NM89N:Device_WateringD2_Limited	0.23
g1_LinePBI17N007M0CM0NM0NM010NM8N:Device_WateringD2_Limited	-0.03
g1_LinePBI17N013M0CM0NM0NM010NM8N:Device_WateringD2_Limited	0.19
g1_LinePBI17N016M0CM0NM0NM010NM27N:Device_WateringD2_Limited	0.25
g1_LinePBI19N007M0NM3N:Device_WateringD2_Limited	0.16
g1_LinePBIC15034M0CM24NM010NM1NM0N:Device_WateringD2_Limited	0.08
g1_LineROCKSTAR:Device_WateringD2_Limited	0.16
g1_LineSUNCHASER:Device_WateringD2_Limited	0.18
g1_Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	0.04
k_Intercept	2.03
k_LinePBI17N002M0CM0NM0NM010NM16N	2.12
k_LinePBI17N006M0NM0NM89N	1.13
k_LinePBI17N007M0CM0NM0NM010NM8N	0.50
k_LinePBI17N013M0CM0NM0NM010NM8N	0.15
k_LinePBI17N016M0CM0NM0NM010NM27N	2.15
k_LinePBI19N007M0NM3N	4.68
k_LinePBIC15034M0CM24NM010NM1NM0N	5.22
k_LineROCKSTAR	3.11
k_LineSUNCHASER	3.98
lambda_Intercept	5.47
	Est.Error
g0_Intercept	0.01
g0_LinePBI17N002M0CM0NM0NM010NM16N	0.01
g0_LinePBI17N006M0NM0NM89N	0.01
g0_LinePBI17N007M0CM0NM0NM010NM8N	0.01
g0_LinePBI17N013M0CM0NM0NM010NM8N	0.01
g0_LinePBI17N016M0CM0NM0NM010NM27N	0.01
g0_LinePBI19N007M0NM3N	0.01
g0_LinePBIC15034M0CM24NM010NM1NM0N	0.01
g0_LineROCKSTAR	0.01
g0_LineSUNCHASER	0.01
g0_Device_WateringD2_Limited	0.02
g0_Room_TemperatureCER8_Ambient	0.02
g0_LinePBI17N002M0CM0NM0NM010NM16N:Device_WateringD2_Limited	0.02
g0_LinePBI17N006M0NM0NM89N:Device_WateringD2_Limited	0.02
g0_LinePBI17N007M0CM0NM0NM010NM8N:Device_WateringD2_Limited	0.02
g0_LinePBI17N013M0CM0NM0NM010NM8N:Device_WateringD2_Limited	0.02
g0_LinePBI17N016M0CM0NM0NM010NM27N:Device_WateringD2_Limited	0.02
g0_LinePBI19N007M0NM3N:Device_WateringD2_Limited	0.02
g0_LinePBIC15034M0CM24NM010NM1NM0N:Device_WateringD2_Limited	0.02

g0_LineROCKSTAR:Device_WateringD2_Limited	0.02
g0_LineSUNCHASER:Device_WateringD2_Limited	0.02
g0_LinePBI17N002M0CM0NM0NM010NM16N:Room_TemperatureCER8_Ambient	0.02
g0_LinePBI17N006M0NM0NM89N:Room_TemperatureCER8_Ambient	0.02
g0_LinePBI17N007M0CM0NM0NM010NM8N:Room_TemperatureCER8_Ambient	0.02
g0_LinePBI17N013M0CM0NM0NM010NM8N:Room_TemperatureCER8_Ambient	0.02
g0_LinePBI17N016M0CM0NM0NM010NM27N:Room_TemperatureCER8_Ambient	0.02
g0_LinePBI19N007M0NM3N:Room_TemperatureCER8_Ambient	0.02
g0_LinePBIC15034M0CM24NM010NM1NM0N:Room_TemperatureCER8_Ambient	0.02
g0_LineROCKSTAR:Room_TemperatureCER8_Ambient	0.02
g0_LineSUNCHASER:Room_TemperatureCER8_Ambient	0.02
g0_Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	0.02
g0_LinePBI17N002M0CM0NM0NM010NM16N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	0.02
g0_LinePBI17N006M0NM0NM89N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	0.02
g0_LinePBI17N007M0CM0NM0NM010NM8N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	0.02
g0_LinePBI17N013M0CM0NM0NM010NM8N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	0.02
g0_LinePBI17N016M0CM0NM0NM010NM27N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	0.02
g0_LinePBI19N007M0NM3N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	0.02
g0_LinePBIC15034M0CM24NM010NM1NM0N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	0.02
g0_LineROCKSTAR:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	0.02
g0_LineSUNCHASER:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	0.02
g1_Intercept	0.05
g1_LinePBI17N002M0CM0NM0NM010NM16N	0.04
g1_LinePBI17N006M0NM0NM89N	0.04
g1_LinePBI17N007M0CM0NM0NM010NM8N	0.04
g1_LinePBI17N013M0CM0NM0NM010NM8N	0.04
g1_LinePBI17N016M0CM0NM0NM010NM27N	0.04
g1_LinePBI19N007M0NM3N	0.04
g1_LinePBIC15034M0CM24NM010NM1NM0N	0.04
g1_LineROCKSTAR	0.04
g1_LineSUNCHASER	0.04
g1_Device_WateringD2_Limited	0.07
g1_Room_TemperatureCER8_Ambient	0.06
g1_LinePBI17N002M0CM0NM0NM010NM16N:Device_WateringD2_Limited	0.06
g1_LinePBI17N006M0NM0NM89N:Device_WateringD2_Limited	0.05
g1_LinePBI17N007M0CM0NM0NM010NM8N:Device_WateringD2_Limited	0.06
g1_LinePBI17N013M0CM0NM0NM010NM8N:Device_WateringD2_Limited	0.06
g1_LinePBI17N016M0CM0NM0NM010NM27N:Device_WateringD2_Limited	0.06
g1_LinePBI19N007M0NM3N:Device_WateringD2_Limited	0.05
g1_LinePBIC15034M0CM24NM010NM1NM0N:Device_WateringD2_Limited	0.06
g1_LineROCKSTAR:Device_WateringD2_Limited	0.06
g1_LineSUNCHASER:Device_WateringD2_Limited	0.06
g1_Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	0.08
k_Intercept	0.32
k_LinePBI17N002M0CM0NM0NM010NM16N	0.73
k_LinePBI17N006M0NM0NM89N	0.39
k_LinePBI17N007M0CM0NM0NM010NM8N	0.34
k_LinePBI17N013M0CM0NM0NM010NM8N	0.14
k_LinePBI17N016M0CM0NM0NM010NM27N	0.55
k_LinePBI19N007M0NM3N	0.63
k_LinePBIC15034M0CM24NM010NM1NM0N	0.96
k_LineROCKSTAR	0.57
k_LineSUNCHASER	0.86
lambda_Intercept	0.45

	1-95% CI
g0_Intercept	-0.02
g0_LinePBI17N002M0CM0NM0NM010NM16N	-0.03
g0_LinePBI17N006M0NM0NM89N	-0.03
g0_LinePBI17N007M0CM0NM0NM010NM8N	-0.01
g0_LinePBI17N013M0CM0NM0NM010NM8N	-0.01
g0_LinePBI17N016M0CM0NM0NM010NM27N	-0.04
g0_LinePBI19N007M0NM3N	-0.04
g0_LinePBIC15034M0CM24NM010NM1NM0N	-0.03
g0_LineROCKSTAR	-0.03
g0_LineSUNCHASER	-0.02
g0_Device_WateringD2_Limited	-0.00
g0_Room_TemperatureCER8_Ambient	-0.04
g0_LinePBI17N002M0CM0NM0NM010NM16N:Device_WateringD2_Limited	-0.05
g0_LinePBI17N006M0NM0NM89N:Device_WateringD2_Limited	-0.04
g0_LinePBI17N007M0CM0NM0NM010NM8N:Device_WateringD2_Limited	-0.04
g0_LinePBI17N013M0CM0NM0NM010NM8N:Device_WateringD2_Limited	-0.03
g0_LinePBI17N016M0CM0NM0NM010NM27N:Device_WateringD2_Limited	-0.04
g0_LinePBI19N007M0NM3N:Device_WateringD2_Limited	-0.04
g0_LinePBIC15034M0CM24NM010NM1NM0N:Device_WateringD2_Limited	0.00
g0_LineROCKSTAR:Device_WateringD2_Limited	-0.04
g0_LineSUNCHASER:Device_WateringD2_Limited	0.00
g0_LinePBI17N002M0CM0NM0NM010NM16N:Room_TemperatureCER8_Ambient	-0.03
g0_LinePBI17N006M0NM0NM89N:Room_TemperatureCER8_Ambient	-0.03
g0_LinePBI17N007M0CM0NM0NM010NM8N:Room_TemperatureCER8_Ambient	-0.04
g0_LinePBI17N013M0CM0NM0NM010NM8N:Room_TemperatureCER8_Ambient	-0.02
g0_LinePBI17N016M0CM0NM0NM010NM27N:Room_TemperatureCER8_Ambient	-0.03
g0_LinePBI19N007M0NM3N:Room_TemperatureCER8_Ambient	-0.04
g0_LinePBIC15034M0CM24NM010NM1NM0N:Room_TemperatureCER8_Ambient	-0.02
g0_LineROCKSTAR:Room_TemperatureCER8_Ambient	-0.03
g0_LineSUNCHASER:Room_TemperatureCER8_Ambient	-0.05
g0_Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	-0.03
g0_LinePBI17N002M0CM0NM0NM010NM16N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	-0.04
g0_LinePBI17N006M0NM0NM89N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	-0.03
g0_LinePBI17N007M0CM0NM0NM010NM8N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	-0.04
g0_LinePBI17N013M0CM0NM0NM010NM8N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	-0.03
g0_LinePBI17N016M0CM0NM0NM010NM27N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	-0.04
g0_LinePBI19N007M0NM3N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	-0.04
g0_LinePBIC15034M0CM24NM010NM1NM0N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	-0.03
g0_LineROCKSTAR:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	-0.04
g0_LineSUNCHASER:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	-0.04
g1_Intercept	0.12
g1_LinePBI17N002M0CM0NM0NM010NM16N	-0.07
g1_LinePBI17N006M0NM0NM89N	0.02
g1_LinePBI17N007M0CM0NM0NM010NM8N	-0.00
g1_LinePBI17N013M0CM0NM0NM010NM8N	-0.03
g1_LinePBI17N016M0CM0NM0NM010NM27N	-0.10
g1_LinePBI19N007M0NM3N	0.05
g1_LinePBIC15034M0CM24NM010NM1NM0N	0.02
g1_LineROCKSTAR	0.03
g1_LineSUNCHASER	0.01
g1_Device_WateringD2_Limited	-0.26
g1_Room_TemperatureCER8_Ambient	-0.11
g1_LinePBI17N002M0CM0NM0NM010NM16N:Device_WateringD2_Limited	0.01

g1_LinePBI17N006M0NM0NM89N:Device_WateringD2_Limited	0.12
g1_LinePBI17N007M0CM0NM0NM010NM8N:Device_WateringD2_Limited	-0.13
g1_LinePBI17N013M0CM0NM0NM010NM8N:Device_WateringD2_Limited	0.08
g1_LinePBI17N016M0CM0NM0NM010NM27N:Device_WateringD2_Limited	0.15
g1_LinePBI19N007M0NM3N:Device_WateringD2_Limited	0.06
g1_LinePBIC15034M0CM24NM010NM1NM0N:Device_WateringD2_Limited	-0.03
g1_LineROCKSTAR:Device_WateringD2_Limited	0.05
g1_LineSUNCHASER:Device_WateringD2_Limited	0.07
g1_Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	-0.11
k_Intercept	1.41
k_LinePBI17N002M0CM0NM0NM010NM16N	0.85
k_LinePBI17N006M0NM0NM89N	0.41
k_LinePBI17N007M0CM0NM0NM010NM8N	0.03
k_LinePBI17N013M0CM0NM0NM010NM8N	0.00
k_LinePBI17N016M0CM0NM0NM010NM27N	1.13
k_LinePBI19N007M0NM3N	3.52
k_LinePBIC15034M0CM24NM010NM1NM0N	3.46
k_LineROCKSTAR	2.04
k_LineSUNCHASER	2.44
lambda_Intercept	4.54
	u-95% CI
g0_Intercept	0.03
g0_LinePBI17N002M0CM0NM0NM010NM16N	0.03
g0_LinePBI17N006M0NM0NM89N	0.02
g0_LinePBI17N007M0CM0NM0NM010NM8N	0.05
g0_LinePBI17N013M0CM0NM0NM010NM8N	0.05
g0_LinePBI17N016M0CM0NM0NM010NM27N	0.02
g0_LinePBI19N007M0NM3N	0.01
g0_LinePBIC15034M0CM24NM010NM1NM0N	0.02
g0_LineROCKSTAR	0.02
g0_LineSUNCHASER	0.04
g0_Device_WateringD2_Limited	0.07
g0_Room_TemperatureCER8_Ambient	0.02
g0_LinePBI17N002M0CM0NM0NM010NM16N:Device_WateringD2_Limited	0.02
g0_LinePBI17N006M0NM0NM89N:Device_WateringD2_Limited	0.03
g0_LinePBI17N007M0CM0NM0NM010NM8N:Device_WateringD2_Limited	0.03
g0_LinePBI17N013M0CM0NM0NM010NM8N:Device_WateringD2_Limited	0.04
g0_LinePBI17N016M0CM0NM0NM010NM27N:Device_WateringD2_Limited	0.03
g0_LinePBI19N007M0NM3N:Device_WateringD2_Limited	0.03
g0_LinePBIC15034M0CM24NM010NM1NM0N:Device_WateringD2_Limited	0.07
g0_LineROCKSTAR:Device_WateringD2_Limited	0.03
g0_LineSUNCHASER:Device_WateringD2_Limited	0.07
g0_LinePBI17N002M0CM0NM0NM010NM16N:Room_TemperatureCER8_Ambient	0.04
g0_LinePBI17N006M0NM0NM89N:Room_TemperatureCER8_Ambient	0.04
g0_LinePBI17N007M0CM0NM0NM010NM8N:Room_TemperatureCER8_Ambient	0.02
g0_LinePBI17N013M0CM0NM0NM010NM8N:Room_TemperatureCER8_Ambient	0.04
g0_LinePBI17N016M0CM0NM0NM010NM27N:Room_TemperatureCER8_Ambient	0.03
g0_LinePBI19N007M0NM3N:Room_TemperatureCER8_Ambient	0.02
g0_LinePBIC15034M0CM24NM010NM1NM0N:Room_TemperatureCER8_Ambient	0.04
g0_LineROCKSTAR:Room_TemperatureCER8_Ambient	0.03
g0_LineSUNCHASER:Room_TemperatureCER8_Ambient	0.01
g0_Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	0.04
g0_LinePBI17N002M0CM0NM0NM010NM16N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	0.03
g0_LinePBI17N006M0NM0NM89N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	0.04

g0_LinePBI17N007M0CM0NM0NM010NM8N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	0.03
g0_LinePBI17N013M0CM0NM0NM010NM8N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	0.05
g0_LinePBI17N016M0CM0NM0NM010NM27N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	0.03
g0_LinePBI19N007M0NM3N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	0.03
g0_LinePBIC15034M0CM24NM010NM1NM0N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	0.05
g0_LineROCKSTAR:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	0.03
g0_LineSUNCHASER:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	0.03
g1_Intercept	0.33
g1_LinePBI17N002M0CM0NM0NM010NM16N	0.09
g1_LinePBI17N006M0NM0NM89N	0.18
g1_LinePBI17N007M0CM0NM0NM010NM8N	0.16
g1_LinePBI17N013M0CM0NM0NM010NM8N	0.13
g1_LinePBI17N016M0CM0NM0NM010NM27N	0.07
g1_LinePBI19N007M0NM3N	0.21
g1_LinePBIC15034M0CM24NM010NM1NM0N	0.18
g1_LineROCKSTAR	0.19
g1_LineSUNCHASER	0.18
g1_Device_WateringD2_Limited	0.00
g1_Room_TemperatureCER8_Ambient	0.15
g1_LinePBI17N002M0CM0NM0NM010NM16N:Device_WateringD2_Limited	0.23
g1_LinePBI17N006M0NM0NM89N:Device_WateringD2_Limited	0.34
g1_LinePBI17N007M0CM0NM0NM010NM8N:Device_WateringD2_Limited	0.08
g1_LinePBI17N013M0CM0NM0NM010NM8N:Device_WateringD2_Limited	0.30
g1_LinePBI17N016M0CM0NM0NM010NM27N:Device_WateringD2_Limited	0.36
g1_LinePBI19N007M0NM3N:Device_WateringD2_Limited	0.27
g1_LinePBIC15034M0CM24NM010NM1NM0N:Device_WateringD2_Limited	0.19
g1_LineROCKSTAR:Device_WateringD2_Limited	0.27
g1_LineSUNCHASER:Device_WateringD2_Limited	0.29
g1_Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	0.20
k_Intercept	2.69
k_LinePBI17N002M0CM0NM0NM010NM16N	3.74
k_LinePBI17N006M0NM0NM89N	1.92
k_LinePBI17N007M0CM0NM0NM010NM8N	1.28
k_LinePBI17N013M0CM0NM0NM010NM8N	0.53
k_LinePBI17N016M0CM0NM0NM010NM27N	3.28
k_LinePBI19N007M0NM3N	5.96
k_LinePBIC15034M0CM24NM010NM1NM0N	7.20
k_LineROCKSTAR	4.29
k_LineSUNCHASER	5.79
lambda_Intercept	6.32

	Rhat
g0_Intercept	1.00
g0_LinePBI17N002M0CM0NM0NM010NM16N	1.00
g0_LinePBI17N006M0NM0NM89N	1.00
g0_LinePBI17N007M0CM0NM0NM010NM8N	1.00
g0_LinePBI17N013M0CM0NM0NM010NM8N	1.00
g0_LinePBI17N016M0CM0NM0NM010NM27N	1.00
g0_LinePBI19N007M0NM3N	1.00
g0_LinePBIC15034M0CM24NM010NM1NM0N	1.00
g0_LineROCKSTAR	1.00
g0_LineSUNCHASER	1.00
g0_Device_WateringD2_Limited	1.00
g0_Room_TemperatureCER8_Ambient	1.00
g0_LinePBI17N002M0CM0NM0NM010NM16N:Device_WateringD2_Limited	1.00

g0_LinePBI17N006M0NM0NM89N:Device_WateringD2_Limited	1.00
g0_LinePBI17N007M0CM0NM0NM010NM8N:Device_WateringD2_Limited	1.00
g0_LinePBI17N013M0CM0NM0NM010NM8N:Device_WateringD2_Limited	1.00
g0_LinePBI17N016M0CM0NM0NM010NM27N:Device_WateringD2_Limited	1.00
g0_LinePBI19N007M0NM3N:Device_WateringD2_Limited	1.00
g0_LinePBIC15034M0CM24NM010NM1NM0N:Device_WateringD2_Limited	1.00
g0_LineROCKSTAR:Device_WateringD2_Limited	1.00
g0_LineSUNCHASER:Device_WateringD2_Limited	1.00
g0_LinePBI17N002M0CM0NM0NM010NM16N:Room_TemperatureCER8_Ambient	1.00
g0_LinePBI17N006M0NM0NM89N:Room_TemperatureCER8_Ambient	1.00
g0_LinePBI17N007M0CM0NM0NM010NM8N:Room_TemperatureCER8_Ambient	1.00
g0_LinePBI17N013M0CM0NM0NM010NM8N:Room_TemperatureCER8_Ambient	1.00
g0_LinePBI17N016M0CM0NM0NM010NM27N:Room_TemperatureCER8_Ambient	1.00
g0_LinePBI19N007M0NM3N:Room_TemperatureCER8_Ambient	1.00
g0_LinePBIC15034M0CM24NM010NM1NM0N:Room_TemperatureCER8_Ambient	1.00
g0_LineROCKSTAR:Room_TemperatureCER8_Ambient	1.00
g0_LineSUNCHASER:Room_TemperatureCER8_Ambient	1.00
g0_Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	1.00
g0_LinePBI17N002M0CM0NM0NM010NM16N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	1.00
g0_LinePBI17N006M0NM0NM89N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	1.00
g0_LinePBI17N007M0CM0NM0NM010NM8N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	1.00
g0_LinePBI17N013M0CM0NM0NM010NM8N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	1.00
g0_LinePBI17N016M0CM0NM0NM010NM27N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	1.00
g0_LinePBI19N007M0NM3N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	1.00
g0_LinePBIC15034M0CM24NM010NM1NM0N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	1.00
g0_LineROCKSTAR:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	1.00
g0_LineSUNCHASER:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	1.00
g1_Intercept	1.00
g1_LinePBI17N002M0CM0NM0NM010NM16N	1.00
g1_LinePBI17N006M0NM0NM89N	1.00
g1_LinePBI17N007M0CM0NM0NM010NM8N	1.00
g1_LinePBI17N013M0CM0NM0NM010NM8N	1.00
g1_LinePBI17N016M0CM0NM0NM010NM27N	1.00
g1_LinePBI19N007M0NM3N	1.00
g1_LinePBIC15034M0CM24NM010NM1NM0N	1.00
g1_LineROCKSTAR	1.00
g1_LineSUNCHASER	1.00
g1_Device_WateringD2_Limited	1.00
g1_Room_TemperatureCER8_Ambient	1.00
g1_LinePBI17N002M0CM0NM0NM010NM16N:Device_WateringD2_Limited	1.00
g1_LinePBI17N006M0NM0NM89N:Device_WateringD2_Limited	1.00
g1_LinePBI17N007M0CM0NM0NM010NM8N:Device_WateringD2_Limited	1.00
g1_LinePBI17N013M0CM0NM0NM010NM8N:Device_WateringD2_Limited	1.00
g1_LinePBI17N016M0CM0NM0NM010NM27N:Device_WateringD2_Limited	1.00
g1_LinePBI19N007M0NM3N:Device_WateringD2_Limited	1.00
g1_LinePBIC15034M0CM24NM010NM1NM0N:Device_WateringD2_Limited	1.00
g1_LineROCKSTAR:Device_WateringD2_Limited	1.00
g1_LineSUNCHASER:Device_WateringD2_Limited	1.00
g1_Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	1.00
k_Intercept	1.00
k_LinePBI17N002M0CM0NM0NM010NM16N	1.00
k_LinePBI17N006M0NM0NM89N	1.00
k_LinePBI17N007M0CM0NM0NM010NM8N	1.00
k_LinePBI17N013M0CM0NM0NM010NM8N	1.00

k_LinePBI17N016M0CM0NM0NM010NM27N	1.00
k_LinePBI19N007M0NM3N	1.00
k_LinePBIC15034M0CM24NM010NM1NM0N	1.00
k_LineROCKSTAR	1.00
k_LineSUNCHASER	1.00
lambda_Intercept	1.00
	Bulk_ESS
g0_Intercept	21374
g0_LinePBI17N002M0CM0NM0NM010NM16N	33359
g0_LinePBI17N006M0NM0NM89N	34740
g0_LinePBI17N007M0CM0NM0NM010NM8N	34771
g0_LinePBI17N013M0CM0NM0NM010NM8N	35655
g0_LinePBI17N016M0CM0NM0NM010NM27N	32398
g0_LinePBI19N007M0NM3N	32076
g0_LinePBIC15034M0CM24NM010NM1NM0N	32546
g0_LineROCKSTAR	31874
g0_LineSUNCHASER	32576
g0_Device_WateringD2_Limited	14681
g0_Room_TemperatureCER8_Ambient	23363
g0_LinePBI17N002M0CM0NM0NM010NM16N:Device_WateringD2_Limited	38994
g0_LinePBI17N006M0NM0NM89N:Device_WateringD2_Limited	42742
g0_LinePBI17N007M0CM0NM0NM010NM8N:Device_WateringD2_Limited	40578
g0_LinePBI17N013M0CM0NM0NM010NM8N:Device_WateringD2_Limited	40674
g0_LinePBI17N016M0CM0NM0NM010NM27N:Device_WateringD2_Limited	39445
g0_LinePBI19N007M0NM3N:Device_WateringD2_Limited	36801
g0_LinePBIC15034M0CM24NM010NM1NM0N:Device_WateringD2_Limited	37478
g0_LineROCKSTAR:Device_WateringD2_Limited	38905
g0_LineSUNCHASER:Device_WateringD2_Limited	35010
g0_LinePBI17N002M0CM0NM0NM010NM16N:Room_TemperatureCER8_Ambient	37793
g0_LinePBI17N006M0NM0NM89N:Room_TemperatureCER8_Ambient	40907
g0_LinePBI17N007M0CM0NM0NM010NM8N:Room_TemperatureCER8_Ambient	38269
g0_LinePBI17N013M0CM0NM0NM010NM8N:Room_TemperatureCER8_Ambient	37654
g0_LinePBI17N016M0CM0NM0NM010NM27N:Room_TemperatureCER8_Ambient	37029
g0_LinePBI19N007M0NM3N:Room_TemperatureCER8_Ambient	33867
g0_LinePBIC15034M0CM24NM010NM1NM0N:Room_TemperatureCER8_Ambient	34421
g0_LineROCKSTAR:Room_TemperatureCER8_Ambient	38740
g0_LineSUNCHASER:Room_TemperatureCER8_Ambient	34969
g0_Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	27535
g0_LinePBI17N002M0CM0NM0NM010NM16N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	40641
g0_LinePBI17N006M0NM0NM89N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	41087
g0_LinePBI17N007M0CM0NM0NM010NM8N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	41760
g0_LinePBI17N013M0CM0NM0NM010NM8N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	38157
g0_LinePBI17N016M0CM0NM0NM010NM27N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	40082
g0_LinePBI19N007M0NM3N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	40014
g0_LinePBIC15034M0CM24NM010NM1NM0N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	40476
g0_LineROCKSTAR:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	40652
g0_LineSUNCHASER:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	38947
g1_Intercept	8691
g1_LinePBI17N002M0CM0NM0NM010NM16N	7446
g1_LinePBI17N006M0NM0NM89N	7400
g1_LinePBI17N007M0CM0NM0NM010NM8N	6840
g1_LinePBI17N013M0CM0NM0NM010NM8N	7300
g1_LinePBI17N016M0CM0NM0NM010NM27N	7440
g1_LinePBI19N007M0NM3N	7969

g1_LinePBIC15034M0CM24NM010NM1NM0N	7484
g1_LineROCKSTAR	7247
g1_LineSUNCHASER	7062
g1_Device_WateringD2_Limited	8662
g1_Room_TemperatureCER8_Ambient	10108
g1_LinePBI17N002M0CM0NM0NM010NM16N:Device_WateringD2_Limited	8697
g1_LinePBI17N006M0NM0NM89N:Device_WateringD2_Limited	8407
g1_LinePBI17N007M0CM0NM0NM010NM8N:Device_WateringD2_Limited	8494
g1_LinePBI17N013M0CM0NM0NM010NM8N:Device_WateringD2_Limited	8925
g1_LinePBI17N016M0CM0NM0NM010NM27N:Device_WateringD2_Limited	8555
g1_LinePBI19N007M0NM3N:Device_WateringD2_Limited	8986
g1_LinePBIC15034M0CM24NM010NM1NM0N:Device_WateringD2_Limited	8070
g1_LineROCKSTAR:Device_WateringD2_Limited	8785
g1_LineSUNCHASER:Device_WateringD2_Limited	8505
g1_Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	12646
k_Intercept	18670
k_LinePBI17N002M0CM0NM0NM010NM16N	29382
k_LinePBI17N006M0NM0NM89N	21077
k_LinePBI17N007M0CM0NM0NM010NM8N	18562
k_LinePBI17N013M0CM0NM0NM010NM8N	24341
k_LinePBI17N016M0CM0NM0NM010NM27N	27715
k_LinePBI19N007M0NM3N	29291
k_LinePBIC15034M0CM24NM010NM1NM0N	28533
k_LineROCKSTAR	29309
k_LineSUNCHASER	28394
lambda_Intercept	24884
	Tail_ESS
g0_Intercept	18377
g0_LinePBI17N002M0CM0NM0NM010NM16N	19700
g0_LinePBI17N006M0NM0NM89N	17875
g0_LinePBI17N007M0CM0NM0NM010NM8N	20434
g0_LinePBI17N013M0CM0NM0NM010NM8N	19229
g0_LinePBI17N016M0CM0NM0NM010NM27N	18345
g0_LinePBI19N007M0NM3N	19680
g0_LinePBIC15034M0CM24NM010NM1NM0N	19343
g0_LineROCKSTAR	18439
g0_LineSUNCHASER	18932
g0_Device_WateringD2_Limited	15747
g0_Room_TemperatureCER8_Ambient	17530
g0_LinePBI17N002M0CM0NM0NM010NM16N:Device_WateringD2_Limited	18962
g0_LinePBI17N006M0NM0NM89N:Device_WateringD2_Limited	18068
g0_LinePBI17N007M0CM0NM0NM010NM8N:Device_WateringD2_Limited	18754
g0_LinePBI17N013M0CM0NM0NM010NM8N:Device_WateringD2_Limited	17797
g0_LinePBI17N016M0CM0NM0NM010NM27N:Device_WateringD2_Limited	17987
g0_LinePBI19N007M0NM3N:Device_WateringD2_Limited	17772
g0_LinePBIC15034M0CM24NM010NM1NM0N:Device_WateringD2_Limited	20138
g0_LineROCKSTAR:Device_WateringD2_Limited	18454
g0_LineSUNCHASER:Device_WateringD2_Limited	18616
g0_LinePBI17N002M0CM0NM0NM010NM16N:Room_TemperatureCER8_Ambient	19011
g0_LinePBI17N006M0NM0NM89N:Room_TemperatureCER8_Ambient	18830
g0_LinePBI17N007M0CM0NM0NM010NM8N:Room_TemperatureCER8_Ambient	19380
g0_LinePBI17N013M0CM0NM0NM010NM8N:Room_TemperatureCER8_Ambient	19274
g0_LinePBI17N016M0CM0NM0NM010NM27N:Room_TemperatureCER8_Ambient	18582
g0_LinePBI19N007M0NM3N:Room_TemperatureCER8_Ambient	19088

```
g0_LinePBIC15034M0CM24NM010NM1NM0N:Room_TemperatureCER8_Ambient                                19926
g0_LineROCKSTAR:Room_TemperatureCER8_Ambient                                                  18773
g0_LineSUNCHASER:Room_TemperatureCER8_Ambient                                                 18410
g0_Device_WateringD2_Limited:Room_TemperatureCER8_Ambient                                     18239
g0_LinePBI17N002M0CM0NM0NM010NM16N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient     17788
g0_LinePBI17N006M0NM0NM89N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient             18352
g0_LinePBI17N007M0CM0NM0NM010NM8N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient      18833
g0_LinePBI17N013M0CM0NM0NM010NM8N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient      17839
g0_LinePBI17N016M0CM0NM0NM010NM27N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient     17755
g0_LinePBI19N007M0NM3N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient                 17878
g0_LinePBIC15034M0CM24NM010NM1NM0N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient     17933
g0_LineROCKSTAR:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient                         18475
g0_LineSUNCHASER:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient                        17616
g1_Intercept                                                                                  12643
g1_LinePBI17N002M0CM0NM0NM010NM16N                                                            12161
g1_LinePBI17N006M0NM0NM89N                                                                    10983
g1_LinePBI17N007M0CM0NM0NM010NM8N                                                             11112
g1_LinePBI17N013M0CM0NM0NM010NM8N                                                             12307
g1_LinePBI17N016M0CM0NM0NM010NM27N                                                            12301
g1_LinePBI19N007M0NM3N                                                                        12614
g1_LinePBIC15034M0CM24NM010NM1NM0N                                                            12195
g1_LineROCKSTAR                                                                               12034
g1_LineSUNCHASER                                                                              11638
g1_Device_WateringD2_Limited                                                                  10798
g1_Room_TemperatureCER8_Ambient                                                               13099
g1_LinePBI17N002M0CM0NM0NM010NM16N:Device_WateringD2_Limited                                  13247
g1_LinePBI17N006M0NM0NM89N:Device_WateringD2_Limited                                          12780
g1_LinePBI17N007M0CM0NM0NM010NM8N:Device_WateringD2_Limited                                   13186
g1_LinePBI17N013M0CM0NM0NM010NM8N:Device_WateringD2_Limited                                   13000
g1_LinePBI17N016M0CM0NM0NM010NM27N:Device_WateringD2_Limited                                  11160
g1_LinePBI19N007M0NM3N:Device_WateringD2_Limited                                              13450
g1_LinePBIC15034M0CM24NM010NM1NM0N:Device_WateringD2_Limited                                  12506
g1_LineROCKSTAR:Device_WateringD2_Limited                                                     13301
g1_LineSUNCHASER:Device_WateringD2_Limited                                                    13111
g1_Device_WateringD2_Limited:Room_TemperatureCER8_Ambient                                     14805
k_Intercept                                                                                   17156
k_LinePBI17N002M0CM0NM0NM010NM16N                                                             14264
k_LinePBI17N006M0NM0NM89N                                                                     12115
k_LinePBI17N007M0CM0NM0NM010NM8N                                                              11291
k_LinePBI17N013M0CM0NM0NM010NM8N                                                              12762
k_LinePBI17N016M0CM0NM0NM010NM27N                                                             14969
k_LinePBI19N007M0NM3N                                                                         19257
k_LinePBIC15034M0CM24NM010NM1NM0N                                                             17573
k_LineROCKSTAR                                                                                16698
k_LineSUNCHASER                                                                               17917
lambda_Intercept                                                                              15710

Further Distributional Parameters:
      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sigma     0.08      0.00     0.08     0.09 1.00    31612    18569

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

Build a dataframe of model parameters

```
li600.i.param <- expand.grid(Room_Temperature = levels(li600.on.dat$Room_Temperature),
                             Device_Watering = levels(li600.on.dat$Device_Watering),
                             Line = levels(li600.on.dat$Line))
```

Use the emmeans package to get out the expected parameter estimates, along with HPD (credible intervals)

```
g0.emmeans <- as.data.frame(emmeans(fit_on_final,
                                    ~ Room_Temperature * Device_Watering * Line,
                                    nlpar = "g0"))
g0.emmeans <- g0.emmeans %>% rename_with(~paste0("g0_", .x),
                                         -all_of(c("Room_Temperature",
                                                   "Device_Watering",
                                                   "Line")))
g0.emmeans <- within(g0.emmeans, {
  g0_tabquant <- paste0("'", round(g0_emmean, 3), "\n[",
                        round(g0_lower.HPD, 3), ", ",
                        round(g0_upper.HPD, 3), "]'")

})
li600.i.param <- left_join(li600.i.param, g0.emmeans)
```

```
Joining with `by = join_by(Room_Temperature, Device_Watering, Line)`
```

```
g1.emmeans <- as.data.frame(emmeans(fit_on_final,
                                    ~ Room_Temperature * Device_Watering * Line,
                                    nlpar = "g1"))
g1.emmeans <- g1.emmeans %>% rename_with(~paste0("g1_", .x),
                                         -all_of(c("Room_Temperature",
                                                   "Device_Watering",
                                                   "Line")))
g1.emmeans <- within(g1.emmeans, {
  g1_tabquant <- paste0("'", round(g1_emmean, 3), "\n[",
                        round(g1_lower.HPD, 3), ", ",
                        round(g1_upper.HPD, 3), "]'")

})
li600.i.param <- left_join(li600.i.param, g1.emmeans)
```

```
Joining with `by = join_by(Room_Temperature, Device_Watering, Line)`
```

```
k.emmeans <- as.data.frame(emmeans(fit_on_final,
                                   ~ Line,
                                   nlpar = "k"))
k.emmeans <- k.emmeans %>% rename_with(~paste0("k_", .x),
                                       -all_of(c("Line")))
k.emmeans <- within(k.emmeans, {
  k_tabquant <- paste0("'", round(k_emmean, 3), "\n[",
                        round(k_lower.HPD, 3), ", ",
                        round(k_upper.HPD, 3), "]'")

})
li600.i.param <- left_join(li600.i.param, k.emmeans)
```

```
Joining with `by = join_by(Line)`
```

```
lambda.emmeans <- as.data.frame(emmeans(fit_on_final,
                                        ~ 1,
                                        nlpar = "lambda"))
lambda.emmeans <- lambda.emmeans %>% rename_with(~paste0("lambda_", .x))
lambda.emmeans <- within(lambda.emmeans, {
  lambda_tabquant <- paste0("'", round(lambda_emmean, 3), "\n[",
                            round(lambda_lower.HPD, 3), ", ",
                            round(lambda_upper.HPD, 3), "]'")

})
li600.i.param <- cbind(li600.i.param, data.frame(lambda.emmeans[,2:5]))
```

```
Warning in data.frame(..., check.names = FALSE): row names were found from a short
variable and have been discarded
```

Using these parameters, calculate Slmax

- Slmax looks to be reported in the units mmol mˆ-2 sˆ-2
- This involves multiplying g1 and g0 by 1000 and also k by 60
- I can calculate this using the posterior draws for each parameter and then combine them together

```
g0_draws <- emmeans(fit_on_final,
                    ~ Room_Temperature * Device_Watering * Line,
                    nlpar = "g0") %>%
  gather_emmeans_draws()

g1_draws <- emmeans(fit_on_final,
                    ~ Room_Temperature * Device_Watering * Line,
                    nlpar = "g1") %>%
  gather_emmeans_draws()

k_draws <- emmeans(fit_on_final,
                   ~ Line,
                   nlpar = "k") %>%
  gather_emmeans_draws()

combined_draws <- g0_draws %>%
  rename(g0 = .value) %>%
  left_join(g1_draws %>% rename(g1 = .value),
            by = c(".draw", "Room_Temperature", "Device_Watering", "Line")) %>%
  left_join(k_draws %>% rename(k = .value),
            by = c(".draw", "Line"))

combined_draws <- combined_draws %>%
  mutate(Slmax = (g1*1000 - g0*1000) / (exp(1) * k*60))

summary_slopes <- combined_draws %>%
  group_by(Room_Temperature, Device_Watering, Line) %>%
  median_hdi(Slmax) %>%
  rename(Slmax_emmean = Slmax) %>%
  rename(Slmax_lower.HPD = .lower) %>%
  rename(Slmax_upper.HPD = .upper)
```

```
summary_slopes <- within(summary_slopes, {
  Slmax_tabquant <- paste0("'",round(Slmax_emmean, 3), "\n[",
                           round(Slmax_lower.HPD, 3), ", ",
                           round(Slmax_upper.HPD, 3), "]'")

})

li600.i.param <- left_join(li600.i.param,
                           summary_slopes %>% select(Room_Temperature,
                                                     Device_Watering,
                                                     Line,
                                                     Slmax_emmean,
                                                     Slmax_lower.HPD,
                                                     Slmax_upper.HPD,
                                                     Slmax_tabquant))
```

```
Joining with `by = join_by(Room_Temperature, Device_Watering, Line)`
```

```
li600.i.param
```

```
   Room_Temperature Device_Watering                         Line     g0_emmean
1         CER7_Heat    D1_Unlimited PBI16C009-0C-0N-0N-010N-14N  5.845269e-03
2      CER8_Ambient    D1_Unlimited PBI16C009-0C-0N-0N-010N-14N -4.207326e-03
3         CER7_Heat      D2_Limited PBI16C009-0C-0N-0N-010N-14N  4.117321e-02
4      CER8_Ambient      D2_Limited PBI16C009-0C-0N-0N-010N-14N  3.606240e-02
5         CER7_Heat    D1_Unlimited PBI17N002-0C-0N-0N-010N-16N  5.147822e-03
6      CER8_Ambient    D1_Unlimited PBI17N002-0C-0N-0N-010N-16N  4.373239e-05
7         CER7_Heat      D2_Limited PBI17N002-0C-0N-0N-010N-16N  2.340287e-02
8      CER8_Ambient      D2_Limited PBI17N002-0C-0N-0N-010N-16N  1.657850e-02
9         CER7_Heat    D1_Unlimited         PBI17N006-0N-0N-89N  2.022159e-03
10     CER8_Ambient    D1_Unlimited         PBI17N006-0N-0N-89N -4.254068e-03
11        CER7_Heat      D2_Limited         PBI17N006-0N-0N-89N  3.065694e-02
12     CER8_Ambient      D2_Limited         PBI17N006-0N-0N-89N  3.115375e-02
13        CER7_Heat    D1_Unlimited  PBI17N007-0C-0N-0N-010N-8N  2.572967e-02
14     CER8_Ambient    D1_Unlimited  PBI17N007-0C-0N-0N-010N-8N  5.142990e-03
15        CER7_Heat      D2_Limited  PBI17N007-0C-0N-0N-010N-8N  5.508361e-02
16     CER8_Ambient      D2_Limited  PBI17N007-0C-0N-0N-010N-8N  3.484254e-02
17        CER7_Heat    D1_Unlimited  PBI17N013-0C-0N-0N-010N-8N  2.282354e-02
18     CER8_Ambient    D1_Unlimited  PBI17N013-0C-0N-0N-010N-8N  2.350839e-02
19        CER7_Heat      D2_Limited  PBI17N013-0C-0N-0N-010N-8N  6.633263e-02
20     CER8_Ambient      D2_Limited  PBI17N013-0C-0N-0N-010N-8N  8.178226e-02
21        CER7_Heat    D1_Unlimited PBI17N016-0C-0N-0N-010N-27N -2.719761e-03
22     CER8_Ambient    D1_Unlimited PBI17N016-0C-0N-0N-010N-27N -1.338192e-02
23        CER7_Heat      D2_Limited PBI17N016-0C-0N-0N-010N-27N  2.748097e-02
24     CER8_Ambient      D2_Limited PBI17N016-0C-0N-0N-010N-27N  2.062143e-02
25        CER7_Heat    D1_Unlimited             PBI19N007-0N-3N -6.744578e-03
26     CER8_Ambient    D1_Unlimited             PBI19N007-0N-3N -3.008506e-02
27        CER7_Heat      D2_Limited             PBI19N007-0N-3N  2.248882e-02
28     CER8_Ambient      D2_Limited             PBI19N007-0N-3N  1.765976e-03
29        CER7_Heat    D1_Unlimited PBIC15034-0C-24N-010N-1N-0N -7.120785e-04
30     CER8_Ambient    D1_Unlimited PBIC15034-0C-24N-010N-1N-0N -2.327329e-03
31        CER7_Heat      D2_Limited PBIC15034-0C-24N-010N-1N-0N  7.179241e-02
32     CER8_Ambient      D2_Limited PBIC15034-0C-24N-010N-1N-0N  8.510869e-02
33        CER7_Heat    D1_Unlimited                   ROCKSTAR  3.805325e-04
```

```
34     CER8_Ambient    D1_Unlimited                   ROCKSTAR -8.347621e-03
35        CER7_Heat      D2_Limited                   ROCKSTAR  2.812548e-02
36     CER8_Ambient      D2_Limited                   ROCKSTAR  2.049069e-02
37        CER7_Heat    D1_Unlimited                  SUNCHASER  1.788555e-02
38     CER8_Ambient    D1_Unlimited                  SUNCHASER -1.007259e-02
39        CER7_Heat      D2_Limited                  SUNCHASER  8.794347e-02
40     CER8_Ambient      D2_Limited                  SUNCHASER  5.936952e-02
    g0_lower.HPD g0_upper.HPD               g0_tabquant  g1_emmean g1_lower.HPD
1  -2.349100e-02   0.03398052  '0.006\n[-0.023, 0.034]' 0.23069664   0.12439640
2  -4.173488e-02   0.03364476 '-0.004\n[-0.042, 0.034]' 0.24178339   0.13190921
3   5.255663e-04   0.07808176   '0.041\n[0.001, 0.078]' 0.09041874  -0.02305576
4  -1.138631e-02   0.08239123  '0.036\n[-0.011, 0.082]' 0.14406481   0.02381930
5  -3.391187e-02   0.04363406  '0.005\n[-0.034, 0.044]' 0.23736939   0.11936471
6  -4.809777e-02   0.05128947      '0\n[-0.048, 0.051]' 0.24791451   0.13054914
7  -2.878915e-02   0.07661506  '0.023\n[-0.029, 0.077]' 0.21808503   0.10433045
8  -4.676033e-02   0.07926938  '0.017\n[-0.047, 0.079]' 0.27220020   0.14510005
9  -3.524251e-02   0.03970295   '0.002\n[-0.035, 0.04]' 0.32551614   0.20526026
10 -5.374443e-02   0.04439992 '-0.004\n[-0.054, 0.044]' 0.33589459   0.21953383
11 -2.460914e-02   0.08197569  '0.031\n[-0.025, 0.082]' 0.41342218   0.29735062
12 -3.193933e-02   0.09782705  '0.031\n[-0.032, 0.098]' 0.46730967   0.34032000
13 -1.239220e-02   0.06535468  '0.026\n[-0.012, 0.065]' 0.30803139   0.18881669
14 -4.439833e-02   0.05513988  '0.005\n[-0.044, 0.055]' 0.31897288   0.20449008
15 -1.595874e-06   0.10674007       '0.055\n[0, 0.107]' 0.14159239   0.02305184
16 -2.629857e-02   0.10024382    '0.035\n[-0.026, 0.1]' 0.19606630   0.07089107
17 -1.546988e-02   0.06216641  '0.023\n[-0.015, 0.062]' 0.28234604   0.16317630
18 -2.582076e-02   0.07419390  '0.024\n[-0.026, 0.074]' 0.29368799   0.17416791
19  1.130917e-02   0.12173458   '0.066\n[0.011, 0.122]' 0.33471686   0.21179236
20  1.258632e-02   0.14678616   '0.082\n[0.013, 0.147]' 0.38762588   0.26215061
21 -4.096327e-02   0.03537650 '-0.003\n[-0.041, 0.035]' 0.21424728   0.09677978
22 -6.337786e-02   0.03613112 '-0.013\n[-0.063, 0.036]' 0.22443066   0.11021841
23 -2.389767e-02   0.08019123   '0.027\n[-0.024, 0.08]' 0.32851473   0.20981584
24 -4.443384e-02   0.08136956  '0.021\n[-0.044, 0.081]' 0.38231111   0.25833180
25 -4.639670e-02   0.02958770  '-0.007\n[-0.046, 0.03]' 0.36282795   0.24231178
26 -7.766643e-02   0.01831867  '-0.03\n[-0.078, 0.018]' 0.37323929   0.25598155
27 -3.230700e-02   0.07442298  '0.022\n[-0.032, 0.074]' 0.38549996   0.27182939
28 -6.084581e-02   0.06489176  '0.002\n[-0.061, 0.065]' 0.43932797   0.31305294
29 -3.931113e-02   0.03765950 '-0.001\n[-0.039, 0.038]' 0.33062216   0.21382542
30 -4.831684e-02   0.04808639 '-0.002\n[-0.048, 0.048]' 0.34250026   0.22605040
31  1.845538e-02   0.12701378   '0.072\n[0.018, 0.127]' 0.26972436   0.15330027
32  1.960111e-02   0.14872348    '0.085\n[0.02, 0.149]' 0.32352553   0.20178298
33 -3.867625e-02   0.03812314      '0\n[-0.039, 0.038]' 0.33946563   0.21700195
34 -5.562973e-02   0.04170390 '-0.008\n[-0.056, 0.042]' 0.35075216   0.23247469
35 -2.576963e-02   0.07796517  '0.028\n[-0.026, 0.078]' 0.36250003   0.24439649
36 -4.216734e-02   0.08067039   '0.02\n[-0.042, 0.081]' 0.41604133   0.29384787
37 -2.178779e-02   0.05643281  '0.018\n[-0.022, 0.056]' 0.32417062   0.20394943
38 -5.943074e-02   0.03817372  '-0.01\n[-0.059, 0.038]' 0.33462691   0.21538627
39  3.178168e-02   0.14284380   '0.088\n[0.032, 0.143]' 0.36222523   0.24715914
40 -4.254305e-03   0.12263272  '0.059\n[-0.004, 0.123]' 0.41619740   0.29495841
   g1_upper.HPD            g1_tabquant k_emmean k_lower.HPD k_upper.HPD
1     0.3290268 '0.231\n[0.124, 0.329]' 2.018804    1.400275    2.670959
2     0.3531951 '0.242\n[0.132, 0.353]' 2.018804    1.400275    2.670959
3     0.2034583 '0.09\n[-0.023, 0.203]' 2.018804    1.400275    2.670959
4     0.2753736 '0.144\n[0.024, 0.275]' 2.018804    1.400275    2.670959
5     0.3497541  '0.237\n[0.119, 0.35]' 4.079593    2.771912    5.691556
```

```
6     0.3659392 '0.248\n[0.131, 0.366]' 4.079593    2.771912    5.691556
7     0.3442974 '0.218\n[0.104, 0.344]' 4.079593    2.771912    5.691556
8     0.4064358 '0.272\n[0.145, 0.406]' 4.079593    2.771912    5.691556
9     0.4336722 '0.326\n[0.205, 0.434]' 3.146440    2.379462    4.008405
10    0.4570781  '0.336\n[0.22, 0.457]' 3.146440    2.379462    4.008405
11    0.5331744 '0.413\n[0.297, 0.533]' 3.146440    2.379462    4.008405
12    0.6008451  '0.467\n[0.34, 0.601]' 3.146440    2.379462    4.008405
13    0.4164781 '0.308\n[0.189, 0.416]' 2.511566    1.798587    3.342049
14    0.4388659 '0.319\n[0.204, 0.439]' 2.511566    1.798587    3.342049
15    0.2601385  '0.142\n[0.023, 0.26]' 2.511566    1.798587    3.342049
16    0.3285812 '0.196\n[0.071, 0.329]' 2.511566    1.798587    3.342049
17    0.3891799 '0.282\n[0.163, 0.389]' 2.164836    1.524443    2.858363
18    0.4090314 '0.294\n[0.174, 0.409]' 2.164836    1.524443    2.858363
19    0.4571963 '0.335\n[0.212, 0.457]' 2.164836    1.524443    2.858363
20    0.5268947 '0.388\n[0.262, 0.527]' 2.164836    1.524443    2.858363
21    0.3273444 '0.214\n[0.097, 0.327]' 4.160179    3.094038    5.282430
22    0.3451855  '0.224\n[0.11, 0.345]' 4.160179    3.094038    5.282430
23    0.4467413  '0.329\n[0.21, 0.447]' 4.160179    3.094038    5.282430
24    0.5223268 '0.382\n[0.258, 0.522]' 4.160179    3.094038    5.282430
25    0.4738687 '0.363\n[0.242, 0.474]' 6.679399    5.488080    8.058489
26    0.4950517 '0.373\n[0.256, 0.495]' 6.679399    5.488080    8.058489
27    0.5110698 '0.385\n[0.272, 0.511]' 6.679399    5.488080    8.058489
28    0.5768105 '0.439\n[0.313, 0.577]' 6.679399    5.488080    8.058489
29    0.4471346 '0.331\n[0.214, 0.447]' 7.207846    5.311068    9.142559
30    0.4651559 '0.343\n[0.226, 0.465]' 7.207846    5.311068    9.142559
31    0.3955102  '0.27\n[0.153, 0.396]' 7.207846    5.311068    9.142559
32    0.4677168 '0.324\n[0.202, 0.468]' 7.207846    5.311068    9.142559
33    0.4475802 '0.339\n[0.217, 0.448]' 5.120474    3.988411    6.324647
34    0.4688709 '0.351\n[0.232, 0.469]' 5.120474    3.988411    6.324647
35    0.4828541 '0.363\n[0.244, 0.483]' 5.120474    3.988411    6.324647
36    0.5553542 '0.416\n[0.294, 0.555]' 5.120474    3.988411    6.324647
37    0.4374858 '0.324\n[0.204, 0.437]' 5.952203    4.272297    7.702272
38    0.4525652 '0.335\n[0.215, 0.453]' 5.952203    4.272297    7.702272
39    0.4841496 '0.362\n[0.247, 0.484]' 5.952203    4.272297    7.702272
40    0.5586134 '0.416\n[0.295, 0.559]' 5.952203    4.272297    7.702272
               k_tabquant lambda_emmean lambda_lower.HPD lambda_upper.HPD
1    '2.019\n[1.4, 2.671]'      5.482128         4.544177         6.325284
2    '2.019\n[1.4, 2.671]'      5.482128         4.544177         6.325284
3    '2.019\n[1.4, 2.671]'      5.482128         4.544177         6.325284
4    '2.019\n[1.4, 2.671]'      5.482128         4.544177         6.325284
5   '4.08\n[2.772, 5.692]'      5.482128         4.544177         6.325284
6   '4.08\n[2.772, 5.692]'      5.482128         4.544177         6.325284
7   '4.08\n[2.772, 5.692]'      5.482128         4.544177         6.325284
8   '4.08\n[2.772, 5.692]'      5.482128         4.544177         6.325284
9  '3.146\n[2.379, 4.008]'      5.482128         4.544177         6.325284
10 '3.146\n[2.379, 4.008]'      5.482128         4.544177         6.325284
11 '3.146\n[2.379, 4.008]'      5.482128         4.544177         6.325284
12 '3.146\n[2.379, 4.008]'      5.482128         4.544177         6.325284
13 '2.512\n[1.799, 3.342]'      5.482128         4.544177         6.325284
14 '2.512\n[1.799, 3.342]'      5.482128         4.544177         6.325284
15 '2.512\n[1.799, 3.342]'      5.482128         4.544177         6.325284
16 '2.512\n[1.799, 3.342]'      5.482128         4.544177         6.325284
17 '2.165\n[1.524, 2.858]'      5.482128         4.544177         6.325284
18 '2.165\n[1.524, 2.858]'      5.482128         4.544177         6.325284
```

```
19 '2.165\n[1.524, 2.858]'      5.482128         4.544177         6.325284
20 '2.165\n[1.524, 2.858]'      5.482128         4.544177         6.325284
21  '4.16\n[3.094, 5.282]'      5.482128         4.544177         6.325284
22  '4.16\n[3.094, 5.282]'      5.482128         4.544177         6.325284
23  '4.16\n[3.094, 5.282]'      5.482128         4.544177         6.325284
24  '4.16\n[3.094, 5.282]'      5.482128         4.544177         6.325284
25 '6.679\n[5.488, 8.058]'      5.482128         4.544177         6.325284
26 '6.679\n[5.488, 8.058]'      5.482128         4.544177         6.325284
27 '6.679\n[5.488, 8.058]'      5.482128         4.544177         6.325284
28 '6.679\n[5.488, 8.058]'      5.482128         4.544177         6.325284
29 '7.208\n[5.311, 9.143]'      5.482128         4.544177         6.325284
30 '7.208\n[5.311, 9.143]'      5.482128         4.544177         6.325284
31 '7.208\n[5.311, 9.143]'      5.482128         4.544177         6.325284
32 '7.208\n[5.311, 9.143]'      5.482128         4.544177         6.325284
33  '5.12\n[3.988, 6.325]'      5.482128         4.544177         6.325284
34  '5.12\n[3.988, 6.325]'      5.482128         4.544177         6.325284
35  '5.12\n[3.988, 6.325]'      5.482128         4.544177         6.325284
36  '5.12\n[3.988, 6.325]'      5.482128         4.544177         6.325284
37 '5.952\n[4.272, 7.702]'      5.482128         4.544177         6.325284
38 '5.952\n[4.272, 7.702]'      5.482128         4.544177         6.325284
39 '5.952\n[4.272, 7.702]'      5.482128         4.544177         6.325284
40 '5.952\n[4.272, 7.702]'      5.482128         4.544177         6.325284
           lambda_tabquant Slmax_emmean Slmax_lower.HPD Slmax_upper.HPD
1  '5.482\n[4.544, 6.325]'    0.6794290      0.32013555       1.1074882
2  '5.482\n[4.544, 6.325]'    0.7452611      0.35924306       1.2146830
3  '5.482\n[4.544, 6.325]'    0.1505521     -0.22606947       0.5404211
4  '5.482\n[4.544, 6.325]'    0.3284195     -0.07554399       0.7868139
5  '5.482\n[4.544, 6.325]'    0.3447834      0.13101305       0.5735646
6  '5.482\n[4.544, 6.325]'    0.3690321      0.15469209       0.6236939
7  '5.482\n[4.544, 6.325]'    0.2894785      0.07440144       0.5307413
8  '5.482\n[4.544, 6.325]'    0.3827931      0.14779160       0.6709269
9  '5.482\n[4.544, 6.325]'    0.6270795      0.36332950       0.9346753
10 '5.482\n[4.544, 6.325]'    0.6617134      0.38993857       0.9923271
11 '5.482\n[4.544, 6.325]'    0.7455489      0.44921291       1.0898885
12 '5.482\n[4.544, 6.325]'    0.8518269      0.50281490       1.2366796
13 '5.482\n[4.544, 6.325]'    0.6863542      0.34133341       1.0704760
14 '5.482\n[4.544, 6.325]'    0.7635721      0.40673643       1.1958894
15 '5.482\n[4.544, 6.325]'    0.2122103     -0.10707243       0.5656989
16 '5.482\n[4.544, 6.325]'    0.3937714      0.03854535       0.8049402
17 '5.482\n[4.544, 6.325]'    0.7311242      0.34291114       1.1641792
18 '5.482\n[4.544, 6.325]'    0.7623979      0.36904616       1.2409219
19 '5.482\n[4.544, 6.325]'    0.7588531      0.34439446       1.2601294
20 '5.482\n[4.544, 6.325]'    0.8711278      0.40579368       1.4361795
21 '5.482\n[4.544, 6.325]'    0.3178259      0.13194818       0.5301781
22 '5.482\n[4.544, 6.325]'    0.3496598      0.15086865       0.5708337
23 '5.482\n[4.544, 6.325]'    0.4428561      0.22741155       0.6942774
24 '5.482\n[4.544, 6.325]'    0.5356363      0.28900797       0.8186545
25 '5.482\n[4.544, 6.325]'    0.3374271      0.21658260       0.4676256
26 '5.482\n[4.544, 6.325]'    0.3694043      0.24409359       0.5063941
27 '5.482\n[4.544, 6.325]'    0.3322649      0.20324054       0.4783026
28 '5.482\n[4.544, 6.325]'    0.4026440      0.25979091       0.5716286
29 '5.482\n[4.544, 6.325]'    0.2810495      0.16731590       0.4092626
30 '5.482\n[4.544, 6.325]'    0.2917251      0.17295057       0.4302249
31 '5.482\n[4.544, 6.325]'    0.1677780      0.04950134       0.3073544
```

```
32 '5.482\n[4.544, 6.325]'    0.2035237      0.06583826       0.3564380
33 '5.482\n[4.544, 6.325]'    0.4041522      0.24227871       0.5795692
34 '5.482\n[4.544, 6.325]'    0.4284562      0.25798551       0.6157156
35 '5.482\n[4.544, 6.325]'    0.3996154      0.23059252       0.5957849
36 '5.482\n[4.544, 6.325]'    0.4755147      0.28110286       0.6957406
37 '5.482\n[4.544, 6.325]'    0.3121895      0.16587219       0.4724046
38 '5.482\n[4.544, 6.325]'    0.3529429      0.20278300       0.5289223
39 '5.482\n[4.544, 6.325]'    0.2815461      0.12545637       0.4598956
40 '5.482\n[4.544, 6.325]'    0.3669438      0.19703207       0.5855615
             Slmax_tabquant
1    '0.679\n[0.32, 1.107]'
2   '0.745\n[0.359, 1.215]'
3   '0.151\n[-0.226, 0.54]'
4  '0.328\n[-0.076, 0.787]'
5   '0.345\n[0.131, 0.574]'
6   '0.369\n[0.155, 0.624]'
7   '0.289\n[0.074, 0.531]'
8   '0.383\n[0.148, 0.671]'
9   '0.627\n[0.363, 0.935]'
10   '0.662\n[0.39, 0.992]'
11   '0.746\n[0.449, 1.09]'
12  '0.852\n[0.503, 1.237]'
13   '0.686\n[0.341, 1.07]'
14  '0.764\n[0.407, 1.196]'
15 '0.212\n[-0.107, 0.566]'
16  '0.394\n[0.039, 0.805]'
17  '0.731\n[0.343, 1.164]'
18  '0.762\n[0.369, 1.241]'
19   '0.759\n[0.344, 1.26]'
20  '0.871\n[0.406, 1.436]'
21   '0.318\n[0.132, 0.53]'
22   '0.35\n[0.151, 0.571]'
23  '0.443\n[0.227, 0.694]'
24  '0.536\n[0.289, 0.819]'
25  '0.337\n[0.217, 0.468]'
26  '0.369\n[0.244, 0.506]'
27  '0.332\n[0.203, 0.478]'
28   '0.403\n[0.26, 0.572]'
29  '0.281\n[0.167, 0.409]'
30   '0.292\n[0.173, 0.43]'
31   '0.168\n[0.05, 0.307]'
32  '0.204\n[0.066, 0.356]'
33   '0.404\n[0.242, 0.58]'
34  '0.428\n[0.258, 0.616]'
35    '0.4\n[0.231, 0.596]'
36  '0.476\n[0.281, 0.696]'
37  '0.312\n[0.166, 0.472]'
38  '0.353\n[0.203, 0.529]'
39   '0.282\n[0.125, 0.46]'
40  '0.367\n[0.197, 0.586]'
```

Format this in a nicer way for export to Excel

Keep the columns in the same order as Li6800 results and order first by parameter and then Line

```
li600.i.export <- li600.i.param %>% select(Room_Temperature, Device_Watering, Line,
                                           grep("tabquant", colnames(.))) %>%
  pivot_longer(cols = 4:ncol(.),
               names_to = "params",
               values_to = "values") %>%
  pivot_wider(names_from = c("Room_Temperature", "Device_Watering"),
              values_from = "values")
li600.i.export$params <- factor(li600.i.export$params, levels = c("g0_tabquant",
                                                                  "g1_tabquant",
                                                                  "k_tabquant",
                                                                  "lambda_tabquant",
                                                                  "Slmax_tabquant"))
li600.i.export <- li600.i.export %>% arrange(params, Line) %>%
  select("params", "Line", "CER7_Heat_D2_Limited", "CER8_Ambient_D2_Limited",
         "CER7_Heat_D1_Unlimited", "CER8_Ambient_D1_Unlimited")
li600.i.export
```

```
# A tibble: 50 × 6
   params      Line     CER7_Heat_D2_Limited CER8_Ambient_D2_Limi...[1] CER7_Heat_D1_Unlimited
   <fct>       <fct>    <chr>                <chr>                      <chr>
 1 g0_tabquant PBI16C0... "'0.041\n[0.001, 0.... "'0.036\n[-0.011, 0.0... "'0.006\n[-0.023, 0.0...
 2 g0_tabquant PBI17N0... "'0.023\n[-0.029, 0... "'0.017\n[-0.047, 0.0... "'0.005\n[-0.034, 0.0...
 3 g0_tabquant PBI17N0... "'0.031\n[-0.025, 0... "'0.031\n[-0.032, 0.0... "'0.002\n[-0.035, 0.0...
 4 g0_tabquant PBI17N0... "'0.055\n[0, 0.107]... "'0.035\n[-0.026, 0.1... "'0.026\n[-0.012, 0.0...
 5 g0_tabquant PBI17N0... "'0.066\n[0.011, 0.... "'0.082\n[0.013, 0.14... "'0.023\n[-0.015, 0.0...
 6 g0_tabquant PBI17N0... "'0.027\n[-0.024, 0... "'0.021\n[-0.044, 0.0... "'-0.003\n[-0.041, 0....
 7 g0_tabquant PBI19N0... "'0.022\n[-0.032, 0... "'0.002\n[-0.061, 0.0... "'-0.007\n[-0.046, 0....
 8 g0_tabquant PBIC150... "'0.072\n[0.018, 0.... "'0.085\n[0.02, 0.149... "'-0.001\n[-0.039, 0....
 9 g0_tabquant ROCKSTAR "'0.028\n[-0.026, 0... "'0.02\n[-0.042, 0.08... "'0\n[-0.039, 0.038]'"
10 g0_tabquant SUNCHAS... "'0.088\n[0.032, 0.... "'0.059\n[-0.004, 0.1... "'0.018\n[-0.022, 0.0...
#  40 more rows
#  abbreviated name:CER8_Ambient_D2_Limited
#  1 more variable: CER8_Ambient_D1_Unlimited <chr>
```

Save data and export

```
save.image(saveFile)
write.csv(li600.i.export, "../data/li600.i.param.export.csv", row.names = FALSE,
          quote = TRUE)
```